Race, Class, and Mobility in U.S. Marriage Markets

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Abstract

We study racial-ethnic disparities in marital and economic status by linking American Community Survey respondents born in 1978-87 to their parents' tax records. Conditional on childhood family income (CFI), we find that the average Black non-Hispanic woman obtains 60 percent less partner income than does the average White non-Hispanic woman, driven both by a lower propensity to be partnered and a lower partner CFI rank. These marriage market dynamics account for 85 percent of the observed—and large—gap in intergenerational family income mobility. We also show that mobility gaps are larger, and rates of intermarriage lower, in birth areas with greater CFI inequality and racial-ethnic segregation. We discuss a simple model in which these patterns originate from segmentation of the marriage market along racial-ethnic lines combined with imperfect assortative matching on economic status. We comment on the implications of our findings for policy.

Keyword*:* marriage market, intergenerational mobility, family income, racial inequality, assortative matching, segregation, intermarriage, union formation, stratification economics

JEL Classification: D31, J12, J15

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More than 250 years after the end of slavery and a full generation after the Civil Rights Era, substantial racial-ethnic differences in income and economic position persist in the United States. The relative economic standing of Black Americans improved with the inclusion of non-slave Black labor in the industrial economy, migration out of the South, greater schooling completion, and civil-rights-era programs,[1] but it has stagnated since 1970 (Bound and Freeman 1992; Bayer and Charles 2018). The absence of meaningful convergence over such a long horizon has inspired recent research on intergenerational mobility processes and their contribution to racial-ethnic disparities (e.g. Killewald and Bryan 2018; Pfeffer and Killewald 2018; Akee et al. 2019; Chetty et al. 2020; Collins and Wanamaker 2022; Derenoncourt 2022). Focusing primarily on the personal income distribution, this research has attributed persistent racial-ethnic inequality to low mobility rates in the general population combined with a racial-ethnic gap in upward mobility—particularly for men.

During the post-1970 era in which men's incomes diverged (Binder and Bound 2019), the United States experienced a massive decline in marriage rates (Cherlin 2014; Lundberg et al. 2016). This decline occurred unevenly across racial-ethnic groups, disproportionately impacting women of color (Tucker and Mitchell-Kernan 1995; Wilson 1996; McLanahan and Percheski 2008; Raley et al. 2015; Smock and Schwartz 2020). Contemporary cohorts of young adult women thus face dramatically different probabilities of living with a stably employed partner, based on racial-ethnic identity.[2] This reality raises a question that has received little attention in the literature: how does family income mobility differ across racial-ethnic groups, and what are the marriage market forces underlying these differences? Romantic partners pool resources to invest in education, homeownership, and children; to insure each other against income shocks; and to consume household public goods. Adopting a family income, rather than personal income, perspective can shed new light on the causes and consequences of racial-ethnic inequality—particularly for women, who exhibit smaller racial-ethnic disparities in personal income than men (Chetty et al. 2020).

In this paper, we provide a novel account of racial-ethnic differences in marriage market outcomes and their contributions to intergenerational income mobility for recent birth cohorts of women. To do so, we merge the universe of Internal Revenue Service (IRS) 1040 income tax forms to household data from the 2011-2019 American Community Survey (ACS). The resultant dataset allows us to observe childhood family income (CFI), race-ethnicity, adulthood income, and

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[1] Myrdal (1944) first reported on the persistence of racial disparities in the U.S. economy. For research documenting and explaining partial convergence, see Smith and Welch (1989); Donohue and Heckman (1991); Aaronson and Mazumder (2011); Collins and Wanamaker (2014); Margo (1995, 2016); Derenoncourt and Montialoux (2021).

[2] In 2019, the share of White non-Hispanic women aged 25-54 currently married was 60.0%, whereas the share of Black non-Hispanic women currently married was only 31.1% (calculations based on public-use ACS microdata). Some of these massive differences are offset by higher rates of non-marital cohabitation among Black non-Hispanic women, but cohabitations in the U.S. that do not turn into marriages tend to dissolve within a few years (e.g., Smock and Schwartz, 2020). Among those ages 40-54, an age range in which most stable cohabitations would have turned into marriages, the share of White non-Hispanic women currently married (66.4%) still greatly exceeded the share for Black non-Hispanic women (38.6%). We make efforts in this paper to capture marriage-like relationships that are not yet legal marriages and that are not always identifiable in the ACS, based on our ACS-IRS data linkage described in the next section.

marriage market outcomes for more than one million adults born between 1978 and 1986. Our data linkage also allows us to identify a subset of marriages and cohabitations that are not identifiable in the ACS alone, by observing individuals living in the same ACS household who filed their taxes as a married couple or will file as married in the future. This subset, as a matter of course, is excluded from studies of union formation that use ACS data only.

Our analysis consists of two parts. First, we estimate new statistics across the CFI distribution and racial-ethnic groups on i) union formation propensities (extensive margin), and ii) average partner income, conditional on having formed a union (intensive margin). We also analyze racial-ethnic differences in assortative matching patterns on CFI. We summarize this information with a concept we call expected income from partner *(EIFP)*, which is the product of i) and ii) and describes how much non-wage income individuals belonging to a given CFI-race group can expect from participating in the marriage market. Because expected family income in adulthood depends on *EIFP* as well as personal income dynamics (Altonji et al. 2022), these *EIFP* differences directly translate into inequality in family income mobility, in ways that we quantify for the first time.

This first part of the analysis finds quite large *EIFP* gaps, driven by both the extensive and the intensive margins. We associate the intensive margin gaps with disparate patterns of assortative matching on CFI: the tendency to match within CFI rank is particularly strong at the bottom of the distribution for Black non-Hispanic and American Indian or Alaska Native non-Hispanic couples and relatively weaker for White non-Hispanic and Asian non-Hispanic couples, while the reverse is true at the top of the distribution.[3] Among women from the 10th percentile of the CFI distribution, the gap in *EIFP* between White non-Hispanic women and Black non-Hispanic women is roughly \$16,000 (or 62% of White non-Hispanic women's *EIFP*). At the 90th percentile, the *EIFP* gap rises to \$29,000 (57%). We assess the implications of these disparities for intergenerational family income mobility by computing mobility statistics based on the observed data, as well as a counterfactual dataset in which each woman's observed partner income is replaced with that of a randomly selected White non-Hispanic woman from the same CFI percentile. We document large mobility disparities in the observed data: for example, while 11 percent of White non-Hispanic women from the bottom quintile of the CFI distribution end up in the top quintile of the adulthood family income distribution, only 3 percent of Black non-Hispanic women do. Similarly, while 39 percent of White non-Hispanic women from the top quintile remain in the top quintile, only 16 percent of Black non-Hispanic women do. These disparities are roughly 85% smaller in the counterfactual dataset, highlighting the central role of marriage market dynamics (as opposed to personal income dynamics) in racial-ethnic inequality in women's income mobility.

In the second part of the paper, we set out to explain these disparities in terms of proximate socioeconomic mechanisms. Prior literature has examined cross-sectional disparities by taking a local marriage markets approach, revealing correlations between area-level marriage patterns and variables such as race-specific economic status and sex ratios (e.g., Lichter et al. 1991, 1992; Harris and Ono 2005; Charles and Luoh 2010; Qian et al. 2018; Washington and Walker 2022). Our data

[3] That is, White non-Hispanic women from the bottom of the CFI distribution are likelier to "marry up" in the distribution, while Black non-Hispanic women from the top of the distribution are likelier to "marry down."

on mobility disparities allow us to expand this literature in two ways. First, we classify local marriage markets according to birth location rather than adulthood location. This controls for the possibility that individuals selectively migrate to better marriage markets. Second, the tax data allow us to move beyond standard survey-based variables and to construct detailed measures of area-level CFI inequality and residential segregation by racial-ethnic identity We test how these general exposures to socioeconomic stratification, on top of racial-ethnic disparities in economic status and sex ratios, shape inequality in adulthood marriage market outcomes.

Our area-level regressions reveal striking patterns of inequality. Consistent with the existing literature, we find that women's *EIFP* benefits from increases in the own-race CFI distribution and sex ratio. However, White non-Hispanic women benefit more from these advantages than do Black non-Hispanic women. In addition, we find that conditional on own-race CFI, general exposures to CFI inequality and racial-ethnic segregation in the local area substantially reduce Black non-Hispanic women's *EIFP* relative to White non-Hispanic women's *EIFP*. For example, a one-standard-deviation increase in area-level inequality exerts basically zero effect on White non-Hispanic women's *EIFP*, but reduces Black non-Hispanic women's *EIFP* by 6.1 log points. Similarly, a one-standard-deviation increase in racial-ethnic segregation reduces White non-Hispanic women's *EIFP* by 1.9 log points, but reduces Black non-Hispanic women's *EIFP* by 5.9 log points. We consider a counterfactual simulation in which we equalize race-specific sex ratios, eliminate racial-ethnic segregation, and reduce income inequality to that of Sweden. This counterfactual eliminates 40 percent of the White-Black mobility gap in *EIFP* (i.e., gap in *EIFP* conditional on CFI), and we explain that this estimate is likely a lower bound. It also more than triples the rate of intermarriage.

Finally, we interpret our findings through the lens of economic theories of marriage. We discuss a simple matching model in which individuals are characterized by social identity (White non-Hispanic or Black non-Hispanic) and CFI (Rich or Poor), where White non-Hispanic individuals are richer on average than Black non-Hispanic individuals. We assume a predetermined penalty for a match between individuals of different social identities, and a premium for a match between rich individuals. The former dynamic captures strong tendencies to match along racial-ethnic lines, which could be interpreted as same-race preferences or as structural barriers to intermarriage that depend on variables like neighborhood segregation or other historical features (e.g., Baker 2022). The latter dynamic arises from standard marriage models with household public goods. We show that these assumptions generate a mobility gap so long as matching on economic status is imperfect (e.g., due to idiosyncratic preferences or search frictions). However, they fail to generate the racial-ethnic disparity in assortative matching on CFI that we observe in the data. Some additional dynamic that segments high-CFI White non-Hispanic individuals into their own submarket is necessary to generate the full extent of mobility disparities. The structure of social preferences and search frictions in U.S. marriage markets appear to limit intergenerational convergence of family incomes across ever-salient racial-ethnic boundaries.

The results of our paper indicate that reducing income inequality and residential segregation will promote more inclusive marriage markets and create additional inequality-reducing spillover effects. This could be achieved through housing, neighborhood, and local education policy; criminal justice reform; and race-neutral (e.g., taxes and transfers) and race-conscious (e.g.,

reparations) redistribution.[4] Our account unifies the disparate strands of literature on racial-ethnic differences in income and in family structure and suggests the importance of policy interventions that treat seriously the strongly assortative tendencies of elite White non-Hispanic populations (e.g., McClintock and Murry 2010).

Our analysis of racial-ethnic inequality expands a literature that explores intergenerational marriage market processes and their contributions to socioeconomic inequality in the broader population. We build on the classic study of Chadwick and Solon (2002), which used the Panel Study of Income Dynamics to analyze the contribution of the marriage market to intergenerational family income mobility among women. Other studies have used similar data to study assortative matching on parental income or wealth and implications for intergenerational mobility (e.g., Ermisch et al. 2006, Charles et al. 2013; Choi et al. 2020). Combining survey data and administrative records, we are able to study a larger sample of individuals and couples that permits detailed examination of racial-ethnic disparities and their underlying proximate mechanisms via spatial analysis. Two related studies use intergenerational administrative data to study assortative matching on wealth in Scandinavian countries without attention to race (Wagner et al. 2020, Fagereng et al. 2022).[5]

The paper is structured as follows. Section I describes our data and our methodology for identifying couples. Section II presents statistics from our national samples concerning union formation, assortative matching, *EIFP*, and family income mobility. Section III describes our local marriage market regressions and presents their results. Section IV concludes the paper by discussing a simple matching framework that we use to interpret our results and the broader context of American marriage markets.

I. Data

We use responses to the 2011-2019 American Community Survey (ACS) as the foundation for our analysis samples. The ACS is the largest household survey in the United States, sampling about 3.5 million addresses each year and eliciting information on a variety of individual-level economic, demographic, social, and housing characteristics. We supplement the ACS with intergenerational information on economic background and contemporary information on marital status from IRS 1040 Individual Income Tax Returns. These forms contain the income of all people in the United

[4] See Lang and Spitzer (2020) for a recent discussion of how racial disparities in one economic market are likely generated by complex interactions and disadvantages across multiple markets, in ways not easily captured by standard neoclassical economic models.

[5] More generally, our work is related to U.S. studies that develop a connection between marriage market forces and population inequality (e.g. Western, Bloome and Percheski 2008; Schwartz 2010; Greenwood et al. 2014; Bloome 2017; Eika et al. 2019; Ciscato and Weber 2020; Gonalons-Pons et al. 2021; Binder 2021).

States who file taxes,[6] along with filing information that allow us to identify parent-child and spousal relationships. We observe this information for tax years 1994-1995 and 1998-2019.

This ACS-IRS data linkage is made possible by the internal data infrastructure at the Census Bureau. To facilitate survey and research operations, administrative and survey data are assigned harmonized personal identifiers called Protected Identification Keys (PIKs) using the Census Bureau's Person Identification Validation System (PVS), as described in Wagner and Layne (2014). This procedure probabilistically matches data to a master reference file by Social Security Number, name, location, and date of birth. The PIK process is nearly comprehensive: in the 2013 ACS, about 93% of individuals were assigned a PIK, and nearly all 1040 records receive a PIK (Bond et al. 2013).

Using this information, we construct two samples: one focal sample of individuals that we use to analyze union formation, and a second sample of matched couples that we use to analyze assortative matching and partner income. Appendix Figure A1 illustrates the birth cohorts and age ranges we consider. We describe the construction of these samples below.

A. Defining Childhood Family Income and the Individual Sample

To define a person's economic background, we take the average of the non-negative Adjusted Gross Income (AGI) values of the adults who claimed this person as a dependent when they were between the ages 10 to 18.[7] We call this concept "childhood family income" (CFI). This concept does not tie children to one unchanging set of caregivers, but rather allows parents to change according to who claims the child as a dependent. This reflects modern instability in many children's living arrangements (Smock and Schwartz 2020). We require at least two potential data years of AGI observations before the child turns 18 to construct CFI. Given that we have access to dependent links on the 1994-1995 and 1998-onward tax forms, this means that the 1978 birth cohort is the earliest one we can analyze.

To construct our individual sample, we merge this CFI information onto the 2011-2019 ACS files. Because a valid PIK is necessary for this merge, we restrict our analysis to ACS respondents who were assigned a PIK. We consider the 1979-1986 birth cohorts, as more recent cohorts were not old enough to be of median union-formation age. Given gender differences in age at union formation, we include men born between 1979 and 1983 (orange and gray cells shown in Figure A1) and women born between 1982 and 1986 (gray and green cells). We observe men at ages 31-35 and women at ages 28-32. Given this relatively young age range, we should not automatically interpret our calculations as lifetime partnering rates or expected partner incomes. However, the

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[6] While tax units with sufficiently low incomes are not required to file taxes, low-income individuals still face a strong incentive to file to receive government benefits such as the Earned Income Tax Credit and Child Tax Credit (Ramnath and Tong 2017).

[7] AGI includes wages, dividends, capital gains, business income, and retirement income. We omit negative AGI values from this average because these are often transitory negative shocks to high income households (e.g., large capital losses) and do not represent the permanent income of parents who claim children on their tax forms. This matches the specification of Chetty et al. (2014), who provide further information.

racial-ethnic disparities we identify at these ages are strongly correlated with lifetime disparities. Moreover, this age range is when many individuals invest in higher education, pay down education debt, save for and purchase homes, and plan for or have children. Understanding union formation and family income dynamics during this age range is of clear importance.

Finally, we focus our analysis on U.S.-born ACS respondents. Non-native-born ACS respondents tend to have higher missing rates of PIK and childhood family income than do native-born respondents. Preliminary analyses suggested that this missingness was not random.[8] Thus, we focus on a target population of U.S.-born individuals, for whom we have representative information on childhood economic circumstances and adulthood marriage market outcomes. This target population should be kept in mind especially when considering results for Hispanic and Asian, non-Hispanic individuals. However, it should not be consequential for our main inferences about disparities between White non-Hispanic and Black non-Hispanic individuals. Our sample size amounts to roughly 1.3 million individuals.

B. Defining Partners in the Couple Sample

The ACS indicates how each person in a household is related to the household reference person. Thus, members of our sample who are the reference person, or the partner of the reference person, can easily be linked to their partners. This process is more challenging in multifamily or multigenerational households. For example, if a 28-year-old woman is living in her parents' home along with her partner, she would be listed as a child of the reference person and her partner would be listed as an "other nonrelative," but no information would exist about how the two individuals relate to each other.

Given this limitation, we supplement ACS relationships with filing status information from the IRS 1040s. For each person sampled in ACS year *t,* we merge on the PIK of the first spouse with whom that individual filed taxes jointly in years *{t,t+1,…t+5}*.[9] We link this person to their IRS spouse if neither they nor their IRS spouse is already linked to an ACS partner, both they and their IRS spouse are at least 15 years old according to their ACS response, and their IRS spouse resides in the person's ACS household. We classify couples as unmarried if either partner responds as unmarried to the individual marital status question that is asked of all ACS adult respondents. This procedure identifies additional marriages in which neither member is the ACS reference person. It also captures additional cohabitations for non-ACS reference persons that will become marriages within five years—consistent with our desire of capturing marriage-like relationships. For both methodological and substantive reasons, we limit our focus to different-sex couples. The ACS

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[8] For example, we found a significantly different marriage rate between the full ACS sample and the ACS sample for which we could observe CFI. Moreover, this disparity remained when we reweighted the sample on observables to adjust for missing CFI, so it is more than just the result of observable sociodemographic differences. However, this problem disappeared when we imposed a US-born restriction.

[9] If a person filed taxes jointly with more than one spouse in the same tax year, we did not consider any of those relationships to be marriages for purposes of identifying IRS spouses. These cases may have arisen from filing errors (the legal requirement when filing income taxes jointly is to file with the person to whom one was legally married on the last day of the tax year) or from record linkage errors.

sample of same-sex couples is not large enough to support separate analyses of smaller racial-ethnic groups by family background, while dramatic differences in the social and legal context of same- and different-sex union formation suggest that separate analyses would be necessary.

Appendix Figure A1 shows the additional couples we identify with the IRS data. Approximately 48% of our individual sample reports being married, and of them, we can link 91% to their spouses using just the ACS. Bringing in the IRS 1040 data raises this fraction to 95%. Panel A shows that we identify 6% more married couples in the bottom quintile of the childhood family income distribution and about 3% more in the top quintile, although 10% of married individuals from the bottom quintile remain unlinked. Because the ACS does not have an individual cohabitation question, we cannot perform the same benchmarking exercise. However, Panel B shows that bringing in the IRS information raises the proportion of individuals participating in a non-marital cohabitation by a full percentage-point (or roughly 9%).

Our couple sample consists of all identified couples where one partner is a member of our focal individual sample, and the other partner belongs to a wider set of age ranges traced out in Figure A1. Specifically, in each survey year, the couple sample includes focal men whose partners are between four years younger and one year older than themselves and focal women whose partners are between one year younger and four years older than themselves. While this age range rules out couples where the partners have very different ages, it ensures that the censoring which does occur is consistent over time and by gender. Our final sample consists of approximately 450,000 couples and comprises roughly 75% of all identified couples.

C. Racial-Ethnic Heterogeneity in our Samples

We focus our analysis on five mutually exclusive racial-ethnic groups: White non-Hispanic, Black non-Hispanic, Hispanic of any race, American Indian or Alaska Native non-Hispanic, and Asian non-Hispanic.[10] We assign ACS respondents to racial-ethnic groups according to how they responded to race and ethnicity items on the ACS. During our sample period, the ACS asked respondents to identify race and Hispanic origin separately. Because of the structure of the survey, racial and ethnic identity may be reported second-hand by the reference person. As a result, our estimates can be interpreted through two lenses: (1) how a person's own identity, or the identity prescribed to them by their household head, factors into their marriage market outcomes, or (2) how they might be racialized by prospective partners. We do not take a stand on these channels and maintain both are important mechanisms informing our results. In Section III, we focus our analysis on marriage market disparities between White non-Hispanic and Black non-Hispanic women.

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[10] These definitions are consistent with the guidance of the Office of Management and Budget (OMB) Statistical Policy Directive No. 15: Standards for Maintaining, Collecting, and Presenting Federal Data on Race and Ethnicity. We define each race group as that race alone and non-Hispanic. This implicitly creates a sixth group, which includes non-Hispanic respondents who identify as Native Hawaiian/Pacific Islander, "some other race", or multiple racial categories. For simplicity, we do not report results for the latter two categories, and we do not analyze Native Hawaiian/Pacific Islander respondents separately due to sample size constraints.

Figure 1 summarizes racial-ethnic heterogeneity in CFI distribution and sets the stage for our ensuing analyses. White non-Hispanic and Asian non-Hispanic individuals tend to be more affluent than average, with far less than 20 percent of these individuals originating from the bottom quintile of the national CFI distribution and far more than 20 percent originating from the top quintile. The reverse is the case for Black non-Hispanic, American Indian or Alaska Native non-Hispanic, and Hispanic individuals. White non-Hispanic individuals are almost 4 times less likely to originate from the bottom quintile of the CFI distribution as Black non-Hispanic individuals, almost 4 times more likely to originate from the top quintile.

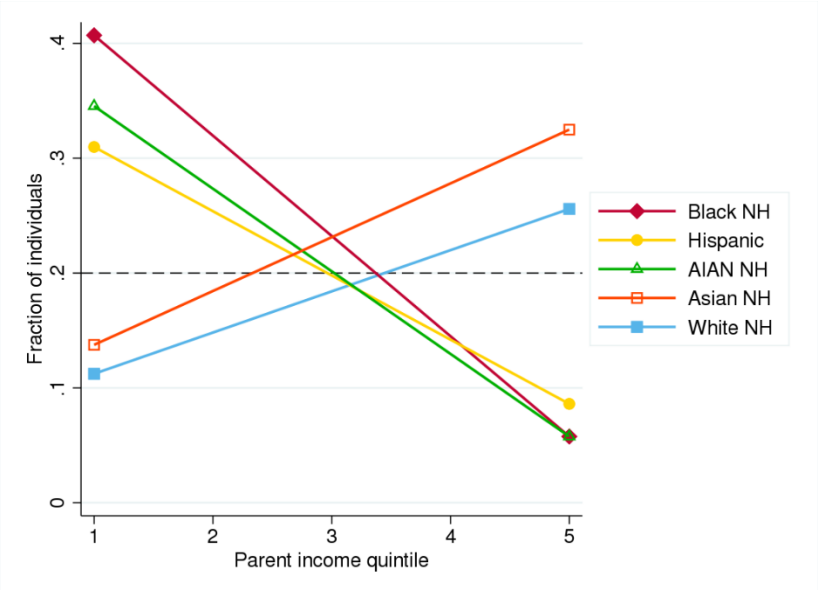


FIGURE 1. CFI BREAKDOWN BY RACIAL-ETHNIC GROUP

Note: See Section I.A for information on individual sample construction and measurement of childhood family income (CFI) rank. "NH" stands for non-Hispanic; "AIAN" stands for American Indian or Alaskan Native.
Source: Individual sample described in Section I.A, which is generated by linking 2011-2019 American Community Survey to 1040 Forms from the Internal Revenue Service.

II. Describing Racial-Ethnic Inequality in Marriage Market Mobility

A. Extensive and Intensive Margins

We begin by using the individual sample to explore how an individual's propensity to participate in a romantic union (the *extensive margin*) varies with their race-ethnicity and CFI percentile. We construct an indicator that takes the value 1 if the individual responds as married to the individual marital status question or if we identify them as a member of a couple, and 0

otherwise.[11] Figure 2 documents substantial variation in union formation propensities for women across racial-ethnic groups, and comparatively less variation across the CFI distribution within a social group. Among women from the 60th percentile of the CFI distribution, roughly 70 percent of White, non-Hispanic women live with a partner, while only 58 percent of Hispanic women and 34 percent of Black non-Hispanic women do. That is, White non-Hispanic women are more than twice as likely to be partnered as Black non-Hispanic women with equivalent economic backgrounds.

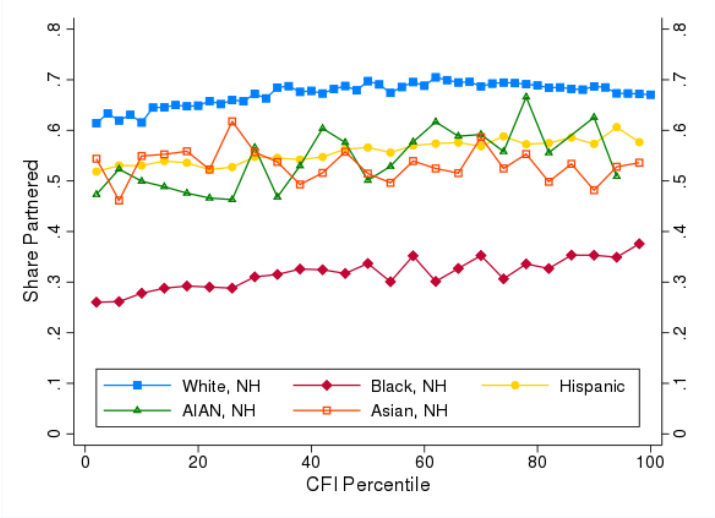


FIGURE 2. SHARE OF WOMEN PARTNERED BY CFI PERCENTILE AND RACE-ETHNICITY

Note: See Section I.A for information on individual sample construction and measurement of childhood family income (CFI) rank. "NH" stands for non-Hispanic; "AIAN" stands for American Indian or Alaskan Native. Our partnered measure captures all marriages, all non-marital cohabitations involving the household head, and all non-marital cohabitations not involving the household head that will become marriages within the next 5 years. See Section I.B for further detail on measurement of couples.
Source: Individual sample described in Section I.A, which is generated by linking 2011-2019 American Community Survey to 1040 Forms from the Internal Revenue Service.

Though an investigation of gender-by-race interaction is outside the scope of this paper, it is worth noting that sex ratios in our individual sample vary across the CFI distribution. High rates of incarceration and mortality among Black men particularly from the bottom of the CFI distribution result in an ACS sample skewed towards more women than men. The implications of this fact are twofold: First, our measure of fraction partnered does not include women with non-cohabiting partners who never marry and hence will understate true rates of romantic unions, particularly for Black non-Hispanic women. Although these relationships are meaningful and

[11] Note that we do not impose the partner age restriction when measuring the extensive margin: we identify all couples we can with our ACS-IRS data linkage, without restricting on birth cohort or age, and then impose the age restriction (Figure A1) later when building the couple sample.

worthy of study, this paper is concerned with a subset of unions that are longer-term, where partners live together, pool resources, and joint financial investments can be made. Incarcerated individuals are unlikely to be able to contribute significantly to these decisions while incarcerated. As a result, we view excluding these relationships as appropriate, rather than a source of bias. Second, the skewed sex ratios in Black non-Hispanic marriage markets result in far fewer same-race potential partners for Black non-Hispanic women than for other racial-ethnic groups. This is a characteristic of marriage markets we investigate directly in Section III, when we examine regional differences in marriage market outcomes.

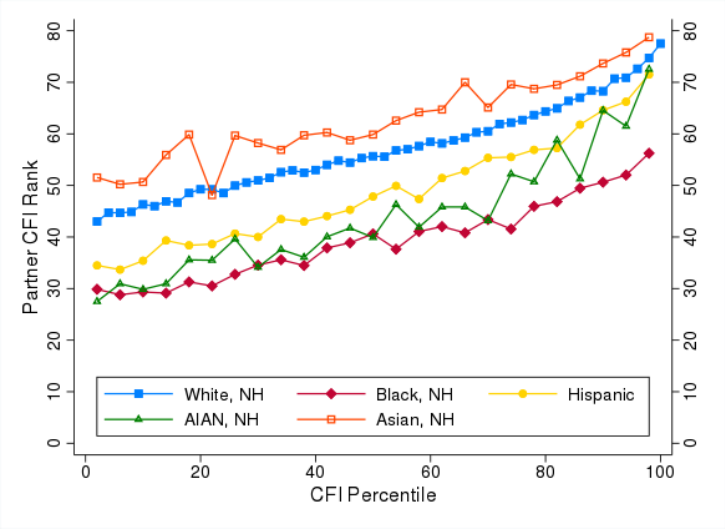


FIGURE 3. PARTNER CFI RANK AS A FUNCTION OF WOMEN'S OWN CFI RANK, BY RACE-ETHNICITY

Note: See Section I.A and I.B for information on couple sample construction and measurement of childhood family income (CFI) rank. "NH" stands for non-Hispanic; "AIAN" stands for American Indian or Alaskan Native. Our sample of couples includes all marriages, all non-marital cohabitations involving the household head, and all non-marital cohabitations not involving the household head that will become marriages within the next 5 years. See Section I.B for further detail.
Source: Couple sample described in Section I.B, which is generated by linking 2011-2019 American Community Survey to 1040 Forms from the Internal Revenue Service.

Figure 3 uses the couple sample to illustrate how women's partner's economic background varies with her own background (the *intensive margin*). The figure shows upward-sloping plots of partner CFI rank against women's own CFI rank for all racial-ethnic groups. This indicates positive assortative matching on CFI, which we explore further in the next subsection. However, the marriage market returns to CFI depend considerably on racial-ethnic identity. For example, we find that the 60th percentile White non-Hispanic woman, on average, partners with a man from the 60th percentile of the CFI distribution. However, an equivalent Hispanic woman partners with a man from the 53rd percentile, while an equivalent Black non-Hispanic woman partners with a man

from the 42nd percentile. The White-Black gap in average partner rank also tends to rise throughout the CFI distribution (apart from the top few percentiles).

B. Assortative Matching on CFI

These disparities in partner CFI rank relate to patterns of assortative matching on CFI, which we analyze in this subsection. We begin by computing a simple criterion $\sigma_{rp}(\tau)$ that measures, for partnered individuals that identify with racial-ethnic group *r* and belong to CFI percentile group *p*, the share of these individuals who partner with someone within τ percentiles of their own rank. Specifically, we classify White non-Hispanic individuals into 50 different percentile groups *p*, and other individuals into 25 different percentile groups.[12] For each partnered woman with exact CFI rank in this group, we assign an indicator $s_{white,p,i}(\tau) = 1$ if *i*'s partner is within τ percentiles of her own exact percentile rank, and 0 otherwise. This yields

$$\sigma_{white,p}(\tau) = \frac{1}{N_{white,p}} \sum_{i \in p} s_{white,p,i}(\tau), \quad (1)$$

where $N_{white,p}$ is the number of partnered White non-Hispanic women belonging to percentile group *p*. We present results based on $\tau = 8$, but they are similar for $\tau = 6$ or 10 as well.[13]

Figure 4 shows considerable variation in the simple criterion across the CFI distribution and for different racial-ethnic groups. CFI sorting is particularly strong at the bottom of the distribution for Black non-Hispanic and Hispanic women, and particularly strong at the top for White non-Hispanic and Asian non-Hispanic women. For example, among individuals from the 10th percentile of the CFI distribution, 40 percent of Black non-Hispanic women partner with someone within 8 percentiles of the 10th percentile, but only 17 percent of White non-Hispanic women do. At the 90th percentile, these numbers reverse: 36 percent of White non-Hispanic women partner within 8 percentiles of 90, but only 17 percent of Black non-Hispanic women do. That is, a Black non-Hispanic woman from the bottom of the CFI distribution is less likely than an equivalent White non-Hispanic woman to "partner up" in the distribution. On the other hand, a Black non-Hispanic woman from the top of the CFI distribution is much likelier than an equivalent White non-Hispanic woman to "partner down" in the distribution. Given that Black non-Hispanic women are likelier than White non-Hispanic women to originate from the bottom of the distribution (Figure 2), this pattern suggests a mobility wedge, whereby social class structure reinforces itself across generations. We provide further evidence of this fact in the next subsection.

The simple criterion clearly illustrates racial-ethnic disparities in marriage market mobility. However, it is not interpretable as a fundamental measure of assortative matching. To understand this point, observe that top-CFI White non-Hispanic individuals have a high propensity to match

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[12] That is, percentile group *p* consists of the (2*p*-1) and 2*p* percentiles of the CFI distribution for White non-Hispanic individuals, and the (4*p*-3) thru (4*p*) percentiles for other groups.

[13] We chose τ=6, 8, or 10 rather than smaller numbers for statistical power and disclosure avoidance issues, particularly for non-White racial-ethnic groups.

with a top-CFI partner. This pattern could occur for two reasons: 1) because individuals match randomly on CFI but assortatively on race-ethnicity, and there are a lot of other top-CFI White non-Hispanic individuals; or 2) because individuals match assortatively on CFI apart from other factors, and these individuals match more assortatively on CFI than do others. Accordingly, we refine Figure 4 by presenting two "formal" assortative matching criteria suggested by Chiappori et al. (2020). Each criterion controls for mechanism 1) and expresses assortativeness generated by mechanism 2). It is informative to explore how both mechanisms determine the mobility wedge.

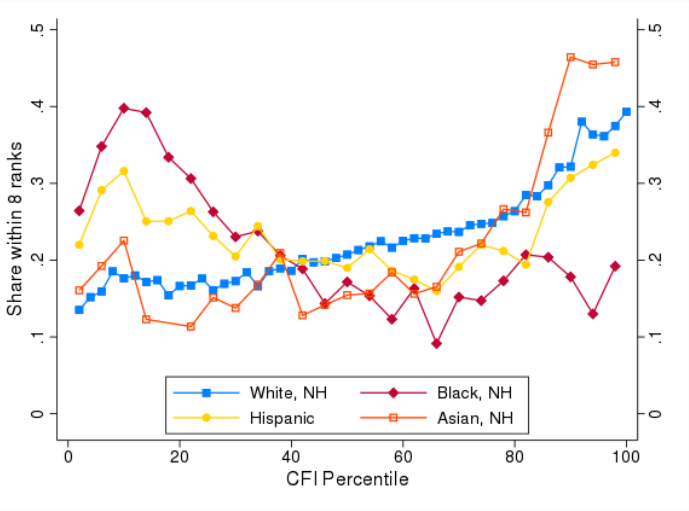


FIGURE 4. SIMPLE ASSORTATIVE MATCHING CRITERION, BY RACE-ETHNICITY AND CFI

Note: Simple criterion records the share of individuals whose partner has a CFI percentile rank within 8 percentiles of their own rank. See Section I for information on sample construction and measurement of childhood family income (CFI) rank. "NH" stands for non-Hispanic. Our sample of couples includes all marriages, all non-marital cohabitations involving the household head, and all non-marital cohabitations not involving the household head that will become marriages within the next 5 years. See Section I.B for further detail on measurement of couples. American Indian / Alaska Native group has been suppressed for disclosure avoidance purposes.
Source: Couple sample described in Section I.B, which is generated by linking 2011-2019 American Community Survey to 1040 Forms from the Internal Revenue Service.

We compute the first formal criterion by regressing an indicator for one partner *i* being within tolerance τ of a given CFI percentile on an indicator for whether their partner *j* is within that same tolerance. More explicitly, for a given racial-ethnic group *r* and CFI percentile *p*, we partition the sample into two categories $c_1 = \{i : P_i \in [p-\tau, p+\tau]\}$; $c_2 = \{i : P_i \notin [p-\tau, p+\tau]\}$. We estimate the following linear probability model separately for each racial-ethnic group and CFI percentile:

$$1\{j \in c_1\} = \alpha_{pr} + \beta_{pr} 1\{i \in c_1\} + \varepsilon_{ipr}$$

This yields the coefficient estimate:

(2) $$\beta_{pr} = Pr(j \in c_1 \mid i \in c_1) - Pr(j \in c_1 \mid i \in c_2)$$

which is the additional probability that a person is from a given CFI percentile group if their partner is also from that group, relative to if their partner is not from that group. This measure takes the simple criterion defined above and subtracts off the probability that a person from outside the CFI percentile group has of forming a union with someone from inside the CFI percentile group. Intuitively, the larger the given group, the larger this offsetting term will be when matching on CFI is random.

The second formal criterion, from Eika et al. (2019), is defined as follows:

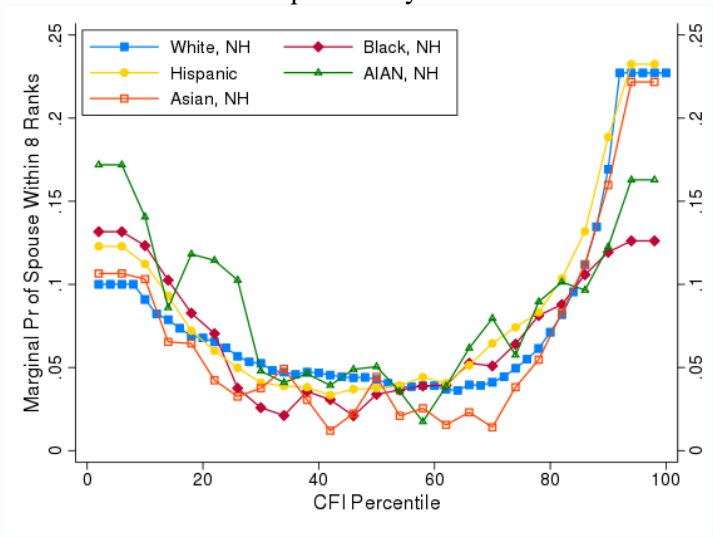
(3) $$EMZ_{pr} = \sum_{n\in\{1,2\}} w_n \cdot \frac{Pr\,(i\in c_n, j\in c_n)}{Pr(i\in c_n)\,Pr(j\in c_n)}$$

Given a partitioning of the sample into two CFI percentile groups, this criterion computes the likelihood that two partners belong to the same group relative to what we would expect if matching on CFI group was random, and then sums the relative likelihoods across the two groups according to weights *w*. Chiappori et al. (2020) demonstrate that this measure is valid for any nonnegative choices of *w* that sum to 1. In fact, it can be shown that if we weight each likelihood ratio by the marginal probability that *i* belongs to the given category (i.e. $w_n = Pr(i \in c_n)$), then $EMZ_{pr} = 1 + \beta_{pr}$.[14] So that our two criteria are not identical, we choose the more complex weighting scheme used by Eika et al. (2019): $w_n = Pr(i \in c_n, j \in c_n)/(Pr(i \in c_1, j \in c_1) + Pr(i \in c_2, j \in c_2))$.

Figure 5 plots each of the two criteria for women, across the CFI distribution and for different racial-ethnic groups. The plots look similar to the simple criterion plot presented above. Both plots continue to show stronger assortative matching on CFI in the tails of the distribution as compared with the middle, with poorer social groups exhibiting the strongest assortative matching at the bottom and richer social groups exhibiting the strongest assortative matching at the top. The weighted EMZ criterion (Panel B), in particular, suggests that Black non-Hispanic and American Indian or Alaska Native non-Hispanic women exhibit much different assortative matching patterns in the tails of the distribution than do White non-Hispanic or Asian non-Hispanic women, in ways that reinforce initial group differences in economic status. Hispanic women, by comparison, exhibit sorting patterns in between these two extremes. These results suggest that the racial disparities in partner CFI rank shown in Figure 3 are driven not just by a tendency for same-race matching, but also by fundamentally different tendencies across racial-ethnic groups to match assortatively on CFI rank. In the next Section, we provide further evidence that the availability of same-race, same-rank partners fundamentally shapes assortative matching patterns—widening mobility gaps beyond what would be predicted by a "neutral" tendency to match within race.

[14] The derivation makes use of the Laws of Conditional Probability and Total Probability and is available on request.

Panel A. Linear probability model coefficient



Panel B. Eika et al. (2019) weighted likelihood ratio

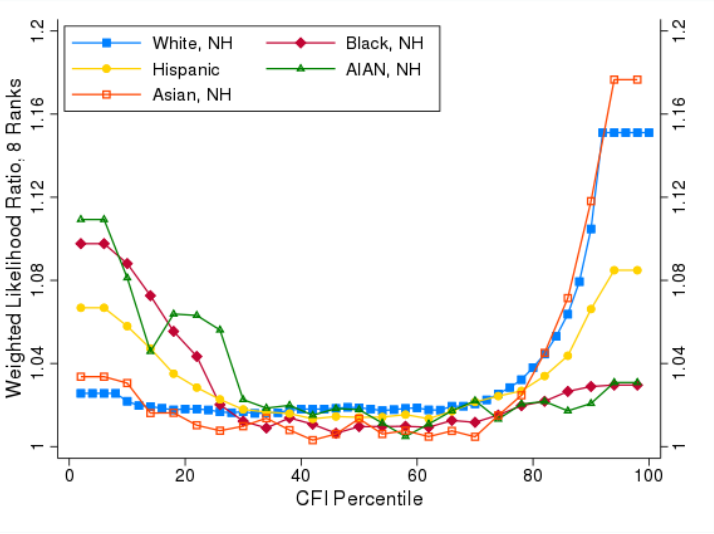


FIGURE 5. FORMAL ASSORTATIVE MATCHING CRITERIA, BY WOMAN PARTNER'S RACE-ETHNICITY AND CFI

Note: See Equations (2) and (3) for descriptions of how each criterion is computed. See Section I for information on sample construction and measurement of childhood family income (CFI) rank. "NH" stands for non-Hispanic; "AIAN" stands for American Indian or Alaskan Native. Our sample of couples includes all marriages, all non-marital cohabitations involving the household head, and all non-marital cohabitations not involving the household head that will become marriages within the next 5 years. See Section Data for further detail on measurement of partnerships.
Source: Couple sample described in Section I.B, which is generated by linking 2011-2019 American Community Survey to 1040 Forms from the Internal Revenue Service.

C. Implications for Inequality in Family Income Mobility

Motivated by these patterns, we combine extensive and intensive margins together into an *expected income from partner* (*EIFP*) statistic. For women identifying with racial-ethnic group r and belonging to CFI percentile p, we write

$$EIFP_{rp} = \underbrace{\Pr(have\ partner \mid r,p)}_{extensive\ margin} \cdot \underbrace{E[partner\ income \mid r,p]}_{intensive\ margin}. \tag{4}$$

This statistic is the product of the probability of living with a romantic partner and the average personal income of partners—measured in adulthood directly from the ACS—and can be considered a measure of the expected marriage market returns for an individual of type rp. As noted in the Introduction, *EIFP* is a direct input into expected adulthood family income; thus, racial-ethnic inequality in *EIFP* directly contributes to racial-ethnic inequality in family income mobility. We provide some illustrative statistics below.

Figure 6 graphs women's *EIFP* by race-ethnicity and CFI percentile. Conditional on the latter, White non-Hispanic and Asian non-Hispanic women tend to have the highest *EIFP*, while American Indian or Alaska Native non-Hispanic and Black non-Hispanic women have the lowest. At the 10th CFI percentile, the gap in *EIFP* between White non-Hispanic women and Hispanic women is roughly $5,000; the *EIFP* gap between White non-Hispanic and Black non-Hispanic women is a full $16,000. (Income is expressed in 2015 U.S. dollars.) Except for the top few percentiles, these gaps tend to rise throughout the CFI distribution. For example, at the 90th percentile, the *EIFP* gap between White non-Hispanic and Hispanic women is roughly $10,000, and the gap between White non-Hispanic and Black non-Hispanic women is roughly $30,000. The White-Black gaps are large relative to the roughly $9,000 gap in individual earned income between the average White non-Hispanic woman and the average Black non-Hispanic woman aged 25-44, observed in 2019 ACS data.

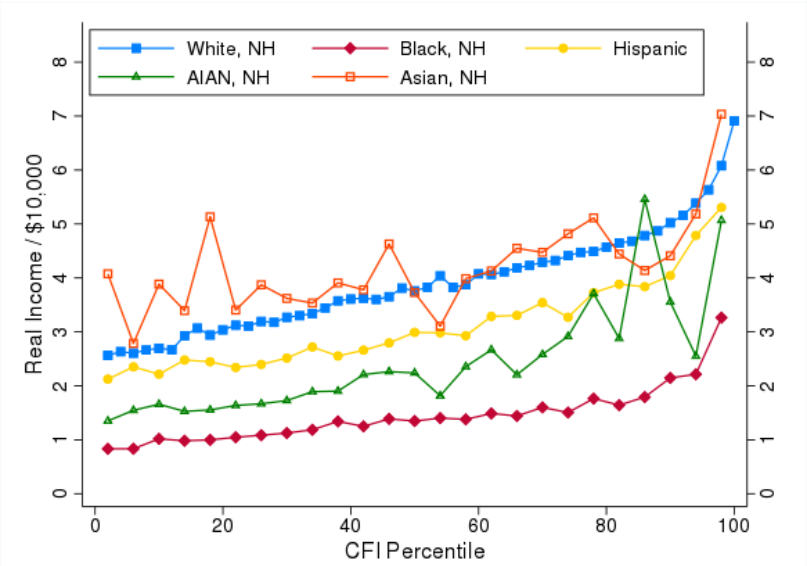


FIGURE 6. WOMEN'S EXPECTED INCOME FROM PARTNER (*EIFP*), BY RACE-ETHNICITY AND CFI PERCENTILE

Note: See Equation (4) for a description of how the *EIFP* statistic is computed. Income is expressed in ten thousands of 2015 U.S. dollars. See Section I for information on sample construction and measurement of childhood family income (CFI) rank. "NH" stands for non-Hispanic; "AIAN" stands for American Indian or Alaskan Native. Our sample of couples includes all marriages, all non-marital cohabitations involving the household head, and all non-marital cohabitations not involving the household head that will become marriages within the next 5 years. See Section I.B for further detail on measurement of couples.
Source: 2011-2019 American Community Survey linked to 1040 Forms from the Internal Revenue Service.

Table 1 maps these differences in CFI-conditional partner income into differences in intergenerational family income mobility. For each woman in our ACS sample, we assign their adulthood family income (AFI) as the sum of their own annual personal income and their partner's (set to zero if no partner is present). We then compute four different mobility statistics that map the distribution of CFI into the distribution of AFI:

i) the share moving into the 2nd-5th quintiles of AFI, conditional on starting in the bottom quintile of CFI,

ii) the share moving to the top quintile conditional on starting at the bottom,
iii) the share remaining in the top quintile conditional on starting there, and
iv) the share moving to the bottom quintile conditional on starting at the top.

Panel A of Table 1 presents observed mobility statistics for women. Consistent with prior work, we find sizable cross-group differences in intergenerational mobility. While 73 percent of White non-Hispanic women starting in the bottom quintile of CFI end up in the 2nd-5th quintiles of AFI, only 57 percent of Black non-Hispanic women do. In addition, while 11 percent of White non-Hispanic women starting in the bottom quintile of CFI end up in the top quintile of AFI, only 3 percent of Black non-Hispanic women do. Similarly large gaps exist for the share remaining in the top quintile (39 percent for White non-Hispanic women and 16 percent for Black non-Hispanic women) and the share moving down to Q1 of AFI from Q5 of CFI (8 percent for White non-Hispanic women versus 20 percent for Black non-Hispanic women).

Panel B presents the same mobility statistics, computed on a counterfactual distribution in which the personal income of each woman from CFI percentile *p* is replaced with the personal income of a randomly-selected White non-Hispanic woman who is also from CFI percentile *p*.[15] Consistent with estimates of modest differences in women's personal income across racial-ethnic groups (e.g. Bayard et al. 1999, Chetty et al. 2020), we find that this counterfactual exerts modest impacts on intergenerational mobility. Compared with the actual data, a slightly larger share of Black non-Hispanic women who start in the top quintile stay there (18 percent versus 16 percent), and a slightly smaller share drop down to the bottom quintile (17 percent versus 20 percent). The same pattern holds for Hispanic women. One exception is that American Indian or Alaska Native non-Hispanic women experience a noteworthy increase in upward mobility and decrease in downward mobility. For example, the share moving out of the bottom quintile rises from 54 to 63 percent, and the share remaining moving from the top quintile down to the bottom decreases from 17 to 12 percent.

Finally, Panel C presents a counterfactual distribution in which women's *partner* incomes are manipulated in the same manner as their personal incomes were in Panel B. In contrast to Panel B, this simulation brings the mobility of Black non-Hispanic and Hispanic women much closer to that of White non-Hispanic women. In this counterfactual scenario, Black non-Hispanic and Hispanic women experience slightly higher rates of moving out of the bottom quintile than do White non-Hispanic women, and they experience only slightly lower rates of moving from the bottom to the top or remaining at the top. Comparing White non-Hispanic and Black non-Hispanic women in Panels A and C, we calculate that replacing observed partner income distributions with those of White non-Hispanic women eliminates 83.6% of the observed disparity in the probability of moving from the bottom to the top quintile, and 89.9% of the observed disparity in the probability of remaining in the top quintile. One note of caution is in order here, which is that the simulation does not take into account the possible endogeneity of women's personal incomes with respect to their partners' incomes (i.e., negative income elasticities of labor supply). However,

[15] To minimize the impact of sampling error, particularly for the American Indian / Alaska Native group, we use a bootstrap procedure in which we randomly simulate 100 different counterfactual distributions, compute mobility statistics for each counterfactual distribution, and then report the average of each group of 100 statistics.

research suggests that these elasticities have become quite small (McClelland and Mok 2012), and the women's incomes are increasingly correlated with their partners' (Gonalons-Pons et al. 2021). The results overall indicate that marriage market dynamics are a key force behind current racial-ethnic disparities in intergenerational mobility.[16]

TABLE 1—INTERGENERATIONAL MOBILITY OF WOMEN'S FAMILY INCOMES

Mobility statistic	White, NH	Black, NH	Hispanic	AIAN, NH
Panel A. Observed Data				
Pr(move out of Q1)	0.73	0.57	0.72	0.54
Pr(move from Q1 to Q5)	0.11	0.03	0.08	0.03
Pr(remain in Q5)	0.39	0.16	0.29	0.22
Pr(move from Q5 to Q1)	0.08	0.20	0.13	0.17
Panel B. Counterfactual: All Groups have White Women's Personal Income				
Pr(move out of Q1)	0.73	0.56	0.69	0.63
Pr(move from Q1 to Q5)	0.11	0.03	0.07	0.05
Pr(remain in Q5)	0.39	0.18	0.33	0.27
Pr(move from Q5 to Q1)	0.08	0.17	0.11	0.12
Panel C. Counterfactual: All Groups Have White Women's Partner Income				
Pr(move out of Q1)	0.67	0.72	0.74	0.66
Pr(move from Q1 to Q5)	0.09	0.08	0.09	0.06
Pr(remain in Q5)	0.37	0.34	0.35	0.30
Pr(move from Q5 to Q1)	0.11	0.10	0.11	0.13

Note: "NH" stands for non-Hispanic; "AIAN" stands for American Indian or Alaskan Native. "Q" refers to quintile. This table maps the distribution of CFI into the distribution of adult family income (AFI), which is the sum of an individual's personal income and their partner's income (set to zero if no partner present). For example, "Pr(move out Q1)" is the conditional probability that a women from the bottom CFI quintile is not in the bottom quintile of AFI. Panel A is the observed data. Panel B (C) replaces observed personal (partner) income for each woman from CFI percentile *p* with the personal (partner) income of a randomly-selected White non-Hispanic woman also from percentile *p*. This process is repeated 100 times; the reported mobility statistics are averaged across these 100 counterfactual distributions.
Source: 2011-2019 American Community Survey linked to 1040 Forms from the Internal Revenue Service.

[16] Appendix Table A1 presents similar counterfactual simulations in which all groups are assigned Black non-Hispanic women's personal or partner income distributions (rather than White non-Hispanic women's). The results are qualitatively similar: changing personal income distributions exerts small effects on racial-ethnic disparities in intergenerational mobility, while changing partner income distributions eliminates the majority of White-Black and White-Hispanic mobility gaps.

III. Proximate Mechanisms of Social Stratification: Evidence from Local Marriage Market Regressions

The previous section described a mobility wedge in the marriage market, whereby a White non-Hispanic individual from a given CFI percentile obtains, on average, a much higher *EIFP* than does an equivalent Black non-Hispanic individual. These partner income disparities almost completely determine disparities in upward mobility in the family income distribution. In this section, we present novel analyses that shed light on the forces driving this mobility wedge.

Our approach in this section exploits variation in channels of socioeconomic stratification across local marriage markets to estimate the effects of these channels on *EIFP*, separately for White non-Hispanic women and for Black non-Hispanic women. We use the estimated race-specific coefficients to examine how differential exposure to marriage market resources (e.g., group differences in CFI), as well as common exposure to socioeconomic stratification (e.g., overall inequality in CFI) shape disparities in marriage market outcomes. Our primary goal is to understand the mobility wedge, i.e., racial-ethnic disparities in *EIFP* after controlling for differences in initial CFI distribution. We focus on White non-Hispanic and Black non-Hispanic women only in this section for two reasons. First, sample size issues limit the reliable computation of local-area marriage market outcomes for most groups of color. Second, the history of anti-Black discrimination in many local jurisdictions renders this analysis particularly well-suited to the study of disparities between White non-Hispanic and Black non-Hispanic individuals.

A. Regression Methodology and Interpretation

We define a local marriage market as a core-based statistical area (CBSA)—groupings of one or more counties that share an urban center and are economically linked through commuting patterns.[17] We assign each woman to a local marriage market according to their *birth* CBSA, defined using the county of birth recorded in the Census Bureau's version of the Social Security Administration Numident file. We choose birth CBSA rather than adulthood CBSA for two reasons: (1) to avoid concerns about selective migration in adulthood that might arise due to marriage market characteristics, and (2) because childhood circumstances comprise the intergenerational processes we seek to measure. Therefore, this choice of geography can be thought of either as a less endogenous proxy for adult marriage market conditions than those conditions themselves, or as measuring direct effects of childhood exposures on women's adulthood marriage market outcomes.

Formally, we regress the average marriage market outcome observed among women of racial-ethnic group r born in CBSA c—denoted as Y_{cr}—on a set of explanatory variables of interest ($\boldsymbol{M_{cr}}$) and other control variables ($\boldsymbol{X}_{cr}$):

$$Y_{cr} = \alpha_r + \rho_r{}'\boldsymbol{M_{cr}} + \delta_r{}'\boldsymbol{X_{cr}} + \varepsilon_{cr}. \tag{5}$$

[17] This concept is smaller than a commuting zone, allowing for more detailed analysis than is possible with public-use microdata.

We estimate this regression separately for White non-Hispanic women and for Black non-Hispanic women, thus allowing the ρ coefficients of interest to vary by race. We consider the following three outcomes: assortative matching on CFI, *EIFP*, and intermarriage. To measure CFI assortativeness, we estimate a linear probability model of a male partner's membership in the top quintile on the woman partner's own membership in that quintile, just as in equation (2). We call the resulting linear probability model coefficient "Q5 sorting". *EIFP* is computed as in equation (4), except at the CBSA-by-race-ethnicity level rather than the race-ethnicity-by-percentile level. Finally, we compute the share of Black non-Hispanic partnered women who have White non-Hispanic partners at the CBSA level. This allows us to assess intermarriage as a potential intermediate channel through which socioeconomic forces generate *EIFP* gaps between White non-Hispanic and Black non-Hispanic women. To improve precision of our estimates, we require at least 20 individual- or couple-level observations within a given CBSA-by-race cell. We also weight each cell by the sum of the ACS person weights contained within it.

For each birth-CBSA-by-racial-ethnic group, we construct several explanatory variables of interest. First, we consider the share of individuals with top-quintile CFI as a measure of group-level initial economic status.[18] This variable allows us to control for initial economic status in *EIFP* regressions to isolate mobility gaps and to examine whether Black non-Hispanic and White non-Hispanic women are rewarded similarly for similar childhood economic statuses. Second, consistent with the preceding literature, we define sex ratios for people living in housing units (i.e., adjusted to exclude individuals in group quarters such as prisons) to examine the impact of sex imbalances arising from differential mortality or incarceration on marriage patterns.

Third, we examine the impact of overall inequality on race-specific marriage market outcomes by defining a CFI inequality measure at the CBSA level. This measure is the natural log of the mean CFI among those from the top quintile, minus the natural log of the mean CFI among those from the bottom four quintiles. This measure effectively describes how many times richer the typical top-quintile person is relative to the typical non-top-quintile person.[19] Fourth, we examine the impact of residential racial segregation by computing a standard dissimilarity index measure (Jahn, et al. 1947). This measure ranges from 0 to 1 and describes the percentage of people that would need to move to a different tract within a CBSA to equalize the share of Black non-Hispanic (or White non-Hispanic) individuals across all tracts within the CBSA.[20] That is, 0 reflects the complete absence of segregation and 1 indicates perfect segregation. To increase precision, we make use of the full universe of IRS 1040 records that match our birth cohort restrictions, rather than the ACS, to construct this index.[21] With these measures of overall social stratification, we can

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[18] Results were similar when we instead defined this measure as the share of individuals with bottom-quintile CFI.

[19] We found similar results when we used the bottom quintile, rather than the top quintile, to partition groups for constructing our log inequality measure.

[20] Formally, the dissimilarity index for CBSA c is defined as $D_c = \frac{1}{2}\sum_{t=1}^{N} \left| \frac{a_t}{A_c} - \frac{b_t}{B_c} \right|$, where a_t is the population count of group Type A in tract t and A_c is the population count of group type A in the CBSA and defined analogously for group type B.

[21] We use address information on the tax records to assign CBSA and tract identifiers. To generate accurate population counts for each racial-ethnic group, we match the tax records to the Census Best Race and Ethnicity file,

examine whether, holding constant a given group's CFI and sex ratio, a more unequal or segregated society results in more segmented markets that benefit White non-Hispanic *EIFP* relative to Black non-Hispanic *EIFP*.

Our control variables include region fixed effects and the natural log of the CBSA population. Fixed effects absorb time-invariant regional marriage market characteristics, such as differences in norms surrounding marriage. Controlling for population allows us to abstract from market dynamics driven by scale. We also control for residential income segregation by computing the same dissimilarity index, except for top-quintile-CFI versus not-top-quintile CFI individuals, instead of according to racial-ethnic categories.

Despite these control variables, one might note that because birth location is not plausibly random, it is difficult to assign a causal interpretation to our estimated relationships between $\boldsymbol{M_{cr}}$ and marriage market outcomes. While we do not dispute this, we contend that the main issue our design faces is not one of econometric bias, but rather one of economic interpretation. For example, suppose that Black non-Hispanic individuals born into more-segregated cities have characteristics (unobserved to the econometrician) that are less rewarded in the marriage market than do Black non-Hispanic individuals born into less-segregated cities. We interpret this extra disadvantage not as a confounding exogenous variable, but rather as a variable endogenous to the forces that produce a segregated society. Such a variable would plausibly respond to an intervention that counteracted segregation-producing forces. Put differently, cross-CBSA inference does not inform us about the short-run effects of a marginal reduction in segregation. However, such an inference is potentially informative about the long-run effects of a large-scale desegregation of American society, allowing for endogenously determined characteristics to adjust in equilibrium. It is exactly this counterfactual scenario that we consider at the end of the Section. Though we do not leverage a specific quasi-experiment, our estimates can be seen as indicative of how the socioeconomic organization of society shapes marriage markets and produces long-run disparities. We view our inferences as akin to those of Chetty et al. (2014) and following work, who interpreted correlations between intergenerational mobility and various political and economic features of the local area.

B. Results

The first two columns of Table 2 present cross-CBSA regression results for Q5 assortative matching. Looking at the first row, we find support for the hypothesis that the availability of same-race, same-rank partners conditions assortative matching. Specifically, we find that a 10 percent increase in the share of Black non-Hispanic individuals from the top quintile increases the Q5 sorting coefficient by .0038, and the effect is statistically significant at the 5 percent level. To provide a sense of magnitude, the standard deviation across CBSAs in the log Q5 share of Black non-Hispanic individuals is roughly 0.6. Thus, a one-standard-deviation increase in Black non-Hispanic CFI raises the Q5 sorting coefficient by .023, or roughly 20 percent of the average Q5

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an internal data product that harmonizes survey and administrative data sources to define single, time-invariant race and ethnicity values for each individual who was assigned a PIK.

sorting coefficient of 0.11.[22] We find qualitatively similar results for White women, although of slightly smaller magnitudes. Looking at the third row of the table, we also find statistically significant effects of prevailing CFI inequality in the CBSA on Q5 sorting. However, the effects point in opposite directions: a 10 percent increase in inequality raises the White Q5 sorting coefficient by .0058, but lowers the Black non-Hispanic Q5 sorting coefficient by .0185. This finding suggests that greater inequality results in a more-differentiated pool of elite White individuals who reap greater relative returns to partnering with each other. On the other hand, greater inequality may lower the access of relatively well-off Black non-Hispanic individuals to elite segments of the marriage market, and hence lower the relative returns to (or feasibility of) finding a partner in the top quintile.[23]

The third and fourth columns of the table report results for *EIFP* (measured on a log scale for ease of interpretation), our main outcome of interest. We find that both White non-Hispanic women and Black non-Hispanic women benefit from higher sex ratios and CFI distributions in the local marriage market. These effects are in line with standard matching theories: if women are scarcer relative to men of their own race or are in a marriage market with a larger share of own-race, high-income men, they are likelier to find a desirable partner. However, Black non-Hispanic women benefit less from these marriage market features. For example, a 10 percent increase in the share of own-race individuals with top-quintile CFI raises White non-Hispanic women's *EIFP* by 1.8 log points, but raises Black non-Hispanic women's *EIFP* by only 0.6 log points. This indicates that the White-Black gap in *EIFP*, and thus in upward marriage market mobility, is particularly large among individuals who originate from the top quintile of the CFI distribution. We also find that a 10 percent increase in the sex ratio raises White non-Hispanic women's *EIFP* by 1.7 log points, but raises Black non-Hispanic women's *EIFP* by a statistically insignificant 0.5 log points.

Moving from race-specific marriage market measures to prevailing measures of socioeconomic stratification (rows 1-2 to rows 3-4 of Table 2), we find that a 10 percent increase in prevailing CFI inequality exerts little effect on White non-Hispanic women's *EIFP* (insignificant increase of 0.3 log points), but substantially lowers Black non-Hispanic women's *EIFP* (significant decrease of 3.1 log points). Similarly, racial-ethnic segregation harms both groups, but especially Black non-Hispanic women: a 10 percentage-point increase in the dissimilarity index reduces White non-Hispanic women's *EIFP* by 1.3 log points, but Black non-Hispanic women's *EIFP* by 3.9 log points. Therefore, racial-ethnic disparities in *EIFP* appear to originate not just from disparities in race-specific marriage market resources, but also from common exposure to an unequal and segmented society.

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[22] The standard deviation in the Q5 sorting coefficient is 0.149, so another way to think about this is that a one-standard-deviation increase in Black non-Hispanic CFI is associated with an increase in Q5 sorting of .0228/.149=0.153 standard deviations.

[23] Our results for White non-Hispanic women, and preliminary analyses in which we pooled racial-ethnic groups together, support the positive relationship between inequality and assortative matching first noted by Fernandez et al. (2005). The differing patterns by average-poor versus average-rich social group, however, add nuance to these findings.

The final column of the table reports results for intermarriage: the CBSA-level probability that a Black non-Hispanic woman partners with a White non-Hispanic man (conditional on her having a partner at all). The results suggest that intermarriage is a mediating channel for the *EIFP* disparities noted above. For example, we find that a 10 percent increase in the share of Black non-Hispanic women with top-quintile CFI increases intermarriage by a statistically significant 0.12 percentage points. Recalling that the standard deviation of this variable is 0.6, a one-standard-deviation increase in the CFI distribution raises intermarriage by 0.72 percentage points. This effect size may not seem large, but it is roughly 12 percent of the mean intermarriage probability in our CBSA sample, which is only 6.1 percent. Similarly, the standard deviation of intermarriage is 3.7 percent, so a one-standard-deviation increase in CFI raises intermarriage by 19.5 percent of a standard deviation. We find that inequality constrains intermarriage: a 10 percent increase in CFI inequality reduces intermarriage by 0.51 percentage points. (A one-standard-deviation increase in inequality reduces intermarriage by 0.82 percentage points, which amounts to 14 percent of the sample mean and 22 percent of a standard deviation.) In addition, racial-ethnic segregation strongly deters intermarriage: a 10-percentage-point increase in the dissimilarity index reduces intermarriage by 1.4 percentage points. (A one-standard-deviation increase in segregation reduces intermarriage by 36 percent of the sample mean and 59 percent of a standard deviation.)

TABLE 2—THE EFFECTS OF SOCIOECONOMIC STRATIFICATION ON WHITE NON-HISPANIC AND BLACK NON-HISPANIC WOMEN'S MARRIAGE MARKET OUTCOMES

	Q5 Sorting		ln(*EIFP*)		Pr(w/ White partner
	White	Black	White	Black	| Black, partnered)
Log share in Q5	0.022*	0.038*	0.179***	0.064**	0.012**
	(.012)	(.020)	(.014)	(.030)	(.005)
Log adj sex ratio	0.033	-.054	0.167***	0.046	-.028*
	(.027)	(.080)	(.035)	(.091)	(.017)
Log Q5 inequality	0.058***	-.185**	0.029	-.307**	-.051**
	(.021)	(.082)	(.036)	(.121)	(.021)
Racial segregation	-.054*	0.031	-.128***	-.393*	-.141***
	(.032)	(.176)	(.048)	(.206)	(.050)
CFI segregation	0.073	0.265	-.049	-.110	0.029
	(.058)	(.301)	(.077)	(0.349)	(.063)
Market size	x	x	x	x	x
Region	x	x	x	x	x
R^2	0.088	0.061	0.358	0.292	0.400
N	850	150	850	150	150

Note: Table presents Ordinary Least Squares regression estimates. "Q" refers to quintile; "adj" stands for "adjusted". Each observation is a CBSA-by-race-ethnicity cell, where individuals are classified according to their birth CBSAs. See Section III.A for information about the construction of explanatory and outcome variables. Additional control variables include "market size"—i.e., the natural log of the total person-weighted CBSA sample size—and indicators for each Census region. Each observation is weighted by its underlying person-weighted sample size. Robust standard errors appear below coefficient estimates in parentheses. Standard statistical significance legend used. *N*s are rounded for disclosure avoidance purposes.

Source: 2011-2019 American Community Survey linked to 1040 Forms from the Internal Revenue Service.

C. Forecasting the Effects of a Large-Scale Reduction in Socioeconomic Stratification

How does the organization of American society shape White-Black gaps in family income? To put the above results in perspective, we consider a counterfactual experiment in which we i) eliminate our observable measures of material disadvantage, by assigning White non-Hispanic women the average CBSA's CFI and sex-ratio characteristics experienced by Black non-Hispanic women; and ii) reduce stratification, by lowering U.S. income inequality to Scandinavian levels and eliminating racial-ethnic segregation. We compute *EIFP* gaps in the observed data and in this counterfactual scenario. As discussed at the beginning of this section, though our cross-CBSA regression coefficients are not causal estimates, they are potentially well-suited to inform us about the effects of long-run, large-scale changes (in which unobserved characteristics that are correlated with the socioeconomic organization of the marriage market endogenously adjust).

First, consider Panel A of Table 3, which reports the average covariate and outcome levels in the observed data. Both White non-Hispanic and Black non-Hispanic women face similar (and high) average levels of CFI inequality. The average log Q5 inequality measure is roughly 1.4, indicating that average CFI level of individuals from the top quintile is $\exp(1.4) \cong 4.1$ times as large as the average CFI level of individuals from the 1st-4th quintiles. There is also a very large gap in CFI levels between White non-Hispanic and Black non-Hispanic individuals: in the typical CBSA, roughly 25 percent of White non-Hispanic individuals originate from the top quintile, whereas only 5 percent of Black non-Hispanic individuals do. Both groups are exposed to highly segregated environments: the average CBSA has a racial-ethnic dissimilarity index of roughly 0.5 in the White non-Hispanic sample and 0.6 in the Black non-Hispanic sample. Finally, the average CBSA contains 0.97 non-institutionalized White non-Hispanic men per White non-Hispanic woman, but only 0.77 non-institutionalized Black non-Hispanic men per Black non-Hispanic woman. This disparity is large, but unsurprising given differential mortality and incarceration rates for Black non-Hispanic men.

Panel B of the table records covariate levels and outcomes in the counterfactual world. The counterfactual scenario eliminates racial-ethnic disparities in CFI distribution and sex ratios, eliminates racial-ethnic segregation, and reduces CFI inequality to Scandinavian levels.[24] We find that the counterfactual experiment closes over half of the 137-log-point *EIFP* gap between White non-Hispanic and Black non-Hispanic women in the average CBSA. That is, the average White non-Hispanic woman goes from having roughly 3.9 times as much partner income as the average Black non-Hispanic woman (in the data) to roughly 1.9 times as much partner income (in the counterfactual world). In level terms, the average partner income gap shrinks from around $30,000 to around $15,000. Moreover, the intermarriage rate more than triples, from around 6 percent in the data to around 19 percent in the counterfactual world. Although substantial partner income gaps remain, and the share of Black non-Hispanic women partnering with White non-Hispanic

[24] The 0.77 value for log Q5 inequality is determined by the following calibration. Under the assumption that family income is distributed lognormally, which is a reasonable approximation to U.S. data, then a 0.77 value for log Q5 inequality is consistent with a Gini coefficient of around 0.26, which is slightly below the Gini for Sweden according to recent OECD data. The reduction in log Q5 inequality from around 1.4 to 0.77 means that the average Q5 family goes from being around 4.1 times as rich as the average Q1-Q4 family to being around 2.2 times as rich.

men remains fairly small, the results suggest that a large-scale reduction of inequality and segregation in America would have substantial spillover effects in the marriage market, and dramatically reduce cross-sectional and intergenerational racial-ethnic disparities in family income.

Two remarks are in order. First, although White non-Hispanic women have a substantial advantage over Black non-Hispanic women in childhood family income, equalizing the CFI distribution across racial-ethnic groups explains only a modest share of our findings. Specifically, reducing the Q5 of share of White non-Hispanic women to that of Black non-Hispanic women reduces the log *EIFP* gap from 1.37 to 1.09. The additional counterfactual reduction from 1.09 to 0.66, reported in the table, occurs conditional on CFI distribution. That is, the counterfactual experiment reduces the mobility wedge in the marriage market by (1.09-0.66)/1.09=40 percent.[25] Second, it is important to emphasize that our estimates only consider CBSA-level disadvantages that are identified by birth location. To the extent that our birth-city-level measures are imperfect proxies for the full extent of Black non-Hispanic disadvantage in marriage markets—due to hyperlocal patterns of disadvantage not reflected in our broader measures, adulthood experiences of disadvantage orthogonal to birth disadvantage, or other dimensions of city-level disadvantage orthogonal to our observed variables—our estimates likely understate the effects of channels of socioeconomic stratification on the mobility wedge (Chetty et al. 2020)

TABLE 3—AVERAGE CBSA-LEVEL COVARIATES AND OUTCOMES FOR WHITE NON-HISPANIC AND BLACK NON-HISPANIC WOMEN

	Panel A. Observed Data			*Panel B. Counterfactual*		
	White	Black	Gap	White	Black	Gap
Covariates						
Log Q5 inequality	1.37	1.48	-0.11	0.77	0.77	0.00
Log share in Q5	-1.40	-2.94	1.54	-2.94	-2.94	0.00
Share in Q5	0.25	0.05	0.19	0.05	0.05	0.00
CFI segregation index	0.37	0.41	-0.04	0.37	0.41	-0.04
Racial segregation index	0.48	0.63	-0.15	0.00	0.00	0.00
Log adjusted sex ratio	-0.03	-0.26	0.23	-0.26	-0.26	0.00
Adjusted sex ratio	0.97	0.77	0.20	0.77	0.77	0.00
Outcomes						
Log *EIFP*	10.61	9.24	1.37	10.37	9.71	0.66
EIFP	40830	10530	30300	31890	16480	15410
Intermarriage		0.06			0.19	

Note: Table presents CBSA-level averages of marriage market covariates and outcomes in the observed data and in the counterfactual scenario. See Section III.A for further information on the construction of the CBSA sample and the lists covariates. Each CBSA observation is weighted by its underlying person-weighted sample size. *EIFP* stands for

[25] We get similar results when we increase the Black non-Hispanic share in Q5 to equal the White non-Hispanic share: the log *EIFP* gap falls from 1.37 to 1.27, and the full counterfactual experiment reduces the gap to 0.79. Thus, the simulation reduces the mobility wedge by (1.27-0.79)/1.27=38 percent.

expected income from partner: see equation (2) for further information on its construction. Intermarriage is probability that a Black non-Hispanic woman partners with a White non-Hispanic man, conditional on having a partner at all. *Source*: 2011-2019 American Community Survey linked to 1040 Forms from the Internal Revenue Service.

IV. Discussion

We have built a novel dataset, by linking American Community Survey responses with tax records, to generate a variety of new findings regarding the intergenerational transmission of family economic status. We have focused our analyses on a comparison between White non-Hispanic and Black non-Hispanic women and have shown the following (conditional on childhood family income, or CFI). First, White non-Hispanic women have roughly double the propensity to live with a romantic partner as do Black non-Hispanic women. Second, White non-Hispanic women partner with much higher-CFI individuals on average than do Black non-Hispanic women. Third, top-CFI White non-Hispanic women are much likelier to match assortatively on CFI than top-CFI Black non-Hispanic women, while non-Hispanic bottom-CFI Black non-Hispanic women are much likelier to match assortatively than bottom-CFI White non-Hispanic women. Fourth, these margins converge to produce a wide mobility wedge, where Black non-Hispanic women can expect 60 percent less partner income from participating in the marriage market as can equivalent White non-Hispanic women. This expected partner income disparity accounts for most racial-ethnic disparities in upward family income mobility across generations. Finally, our local marriage markets analysis has revealed that these disparities are driven by income inequality and racial-ethnic segregation and can thus be viewed as byproducts of a socioeconomically segmented American society.

Given the link we have found between racial-ethnic stratification in local communities and the racial-ethnic gap in mobility through marriage, we doubt whether race-neutral policies would suffice to close the gap. Indeed, as long as race-based inequality and segregation persist, race-neutral interventions might exacerbate the gap. Depending on the structure of the marriage market, a policy which increases the upward mobility of low CFI individuals of all races may result in a wider gap in mobility between Black non-Hispanic and White non-Hispanic women due to the differences in assortativeness we've identified.

To illustrate this point, consider a simple matching environment in which individuals are characterized by either of two social identities (White non-Hispanic or Black non-Hispanic) and either of two economic statuses (Poor or Rich). We abstract from gender and sex-ratio considerations by assuming that both sides of the matching market have identical sizes and distributions of each characteristic. We consider a matching process in which everyone is assigned to one of two submarkets according to a racial-ethnic preference, or a friction that impedes cross-race interactions such as neighborhood segregation. Each submarket then clears according to a centralized matching process.

Table 4 records an equilibrium allocation of this matching process, separately by race. The parameter q_R describes the share of individuals of race group R that are Rich, while β_R is the linear probability model assortative matching index that we have discussed earlier. Let r_R correspond to the share of race group R's partners that are rich. Our working assumption is that $r_B < r_W$: market

segmentation constrains Black non-Hispanic individuals to have an average-poorer set of partners from which to choose. (This could follow from same-race preferences combined with the fact that Black non-Hispanic individuals are poorer than White non-Hispanic individuals, i.e., $q_B < q_W$). Market-clearing constraints imply that the four matching probabilities in each box are between 0 and 1, and sum to 1. These determine the endogenous object $r_R^* = r_R - q_R\beta_R$.

TABLE 4—MATCHING PATTERNS BY RACE, IN TERMS OF MARGINAL CFI DISTRIBUTION AND LINEAR PROBABILITY MODEL ASSORTATIVE MATCHING PARAMETER (β)

Partners of Black individuals

		Poor	Rich	Total
Black individuals	Poor	$(1-q_B)(1-r^*_B)$	$(1-q_B)r^*_B$	$1-q_B$
	Rich	$q_B(1-r^*_B-\beta_B)$	$q_B(r^*_B+\beta_B)$	q_B
	Total	$1-(r^*_B+q_B\beta_B)$	$r^*_B+q_B\beta_B$	

Partners of White individuals

		Poor	Rich	Total
White individuals	Poor	$(1-q_W)(1-r^*_W)$	$(1-q_W)r^*_W$	$1-q_W$
	Rich	$q_W(1-r^*_W-\beta_W)$	$q_W(r^*_W+\beta_W)$	q_W
	Total	$1-(r^*_W+q_W\beta_W)$	$r^*_W+q_W\beta_W$	

Note: Equation (2) defines the linear probability model assortative matching parameter, which is β_R for race group R. The parameter q_R is the share of individuals of race group R that are Rich. The endogenous object $r_R^* = r_R - q_R\beta_R$, where r_R is the share of race group R's partners that are Rich. See text for discussion.

To see how a race-neutral intervention to increase mobility could increase the mobility gap between Black non-Hispanic and White non-Hispanic individuals, imagine that matching is perfectly assortative with respect to economic status. This occurs when $\beta_R = 1$ and $r_R^* = 0$. In this market, there is no mobility gap because there is no mobility for anyone: each Poor (Rich) individual is matched to another Poor (Rich) individual, regardless of race.[26] Now consider an intervention that makes matching random with respect to economic status. This occurs when $\beta_R = 0$ and $r_R^* = r_R$. In this post-intervention market, a mobility gap emerges: all Black non-Hispanic (White non-Hispanic) individuals have an r_B (r_W) probability of matching with a Rich partner, yielding improved mobility for both groups and a racial-ethnic mobility gap of $r_W - r_B$.

Our empirical results also showed that assortative matching on CFI is differentially strong for the Richest White non-Hispanic individuals (relative to the Richest Black non-Hispanic

[26] Though we do not explicitly consider union formation in our simple framework, if this decision depends solely on the economic status of potential partners as we have assumed, then we would also expect identical partnering propensities conditional on own economic status as well.

individuals) and the Poorest Black non-Hispanic individuals (relative to the Poorest White non-Hispanic individuals). Although our simplified framework cannot speak to both disparities as that would require a third "middle" CFI category, we can examine how the greater tendency of Rich White non-Hispanic individuals to match assortatively relative to Rich Black non-Hispanic individuals widens the mobility gap for Rich individuals beyond the "race-neutral sorting" discussion above. Maintain our assumption of racial-ethnic segmentation and assume that Black non-Hispanic individuals match randomly with respect to CFI, while White non-Hispanic individuals match perfectly assortatively. Based on the above discussion, it follows that Rich Black non-Hispanic individuals have an r_B probability of matching with a Rich partner, while Rich White non-Hispanic individuals have a probability of 1, yielding a wider mobility gap of $1 - r_B$.

These examples show how racial-ethnic inequality together with racially segmented marriage markets can create tradeoffs between increasing mobility and reducing the racial-ethnic mobility gap. These tradeoffs arise because, in racially segmented marriage markets, mobility increases the extent to which an individual's marriage market fortunes rise and fall with the economic status of their race group. American schooling is a highly relevant example. Given that neighborhood income segregation is less stark than racial-ethnic segregation, public schooling increases the mixing of same-race individuals across economic lines, thus promoting within-group economic mobility. However, the severe extent of racial-ethnic segregation in the public school system implies that preferential egalitarianism towards the majority racial-ethnic group may widen racial-ethnic mobility gaps in ways that spill over to the marriage market. In a similar vein, improving low-income students' access to affluent, predominantly White private schools might reduce assortative matching of Rich White individuals and thus lower racial-ethnic mobility gaps at the top of the CFI distribution. However, so long as these low-income students are themselves predominantly White, this will occur at the expense of a higher racial-ethnic mobility gap at the bottom of the CFI distribution. Policies that engage youths with a more economically diverse set of peers and experiences without addressing racial-ethnic segregation as well might improve one metric of inequality at the expense of another.

Reinforcing this discussion, our CBSA-level regressions show how racial-ethnic disparities in marriage-market outcomes between Black non-Hispanic and White non-Hispanic women are amplified in communities with greater racial-ethnic segregation. These patterns reflect the legacy of devaluation and discrimination against Black non-Hispanic people in the United States, and one might hope they are a fading legacy of the past. As the social and economic positions of Black non-Hispanic and White non-Hispanic Americans converge, preferences for same-race unions and search frictions that impede cross-race partnering may naturally weaken. But as stratification economists have argued (Darity, 2022; Chelwa, Hamilton, and Stewart, 2022), the persistence and magnitude of racial-ethnic disparities in market outcomes may reflect ongoing efforts of dominant groups to shape markets to their advantage. Applied to marriage markets, this alternative view suggests that the conditions of union formation are a tool that affluent groups can use to consolidate resources and to maintain their dominant economic position. If this account is the correct one, then it suggests the importance of policy interventions that treat seriously the exclusionary tendencies of elite White non-Hispanic populations.

Overall, our results suggest that policies which decrease the segmentation of markets by reducing income inequality and racial-ethnic segregation will promote more inclusive marriage

markets and create additional inequality-reducing spillover effects. We have also identified the role that suppressed earnings of Black non-Hispanic men relative to White non-Hispanic men plays in perpetuating mobility gaps in family income for women. Interventions which improve the economic prospects of Black non-Hispanic men specifically will result in higher adulthood family income for the women they partner with. However, given the growing diversification of household structure such as single-parent, extended family, and non-family households (Cherlin 2010), we may also consider policy interventions which improve social and economic outcomes for these types of households as well. This could be achieved through housing, neighborhood, and local education policy; criminal justice reform; and race-neutral (e.g., taxes and transfers) and race-conscious (e.g., reparations) redistribution.

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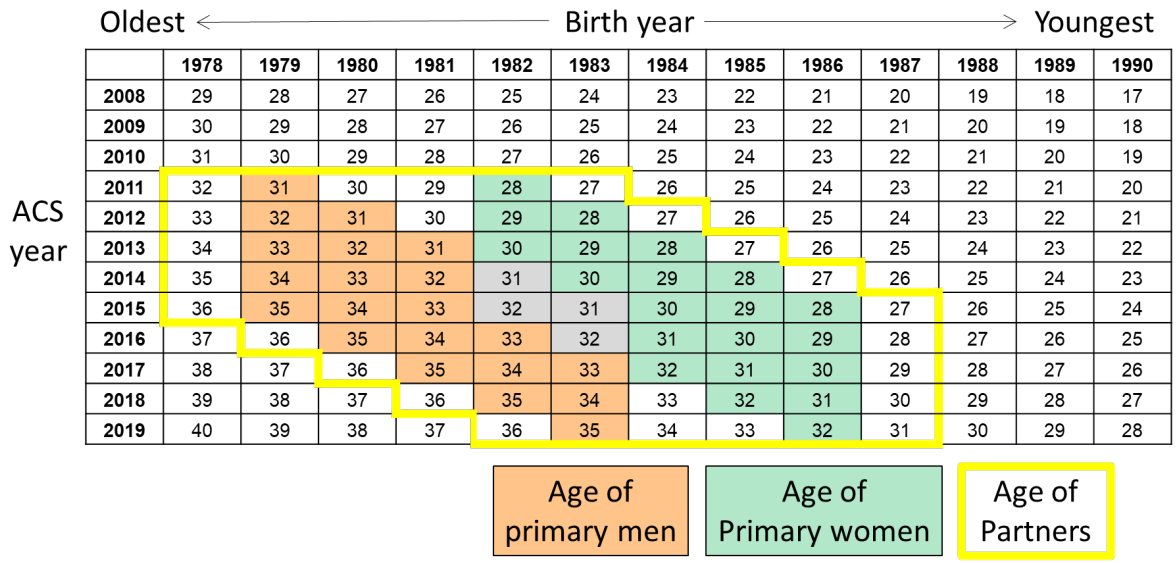
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Appendix Figures and Tables



	1978	1979	1980	1981	1982	1983	1984	1985	1986	1987	1988	1989	1990
2008	29	28	27	26	25	24	23	22	21	20	19	18	17
2009	30	29	28	27	26	25	24	23	22	21	20	19	18
2010	31	30	29	28	27	26	25	24	23	22	21	20	19
2011	32	31	30	29	28	27	26	25	24	23	22	21	20
2012	33	32	31	30	29	28	27	26	25	24	23	22	21
2013	34	33	32	31	30	29	28	27	26	25	24	23	22
2014	35	34	33	32	31	30	29	28	27	26	25	24	23
2015	36	35	34	33	32	31	30	29	28	27	26	25	24
2016	37	36	35	34	33	32	31	30	29	28	27	26	25
2017	38	37	36	35	34	33	32	31	30	29	28	27	26
2018	39	38	37	36	35	34	33	32	31	30	29	28	27
2019	40	39	38	37	36	35	34	33	32	31	30	29	28

FIGURE A1. SAMPLE CONSTRUCTION AND AGE RESTRICTIONS

Note: Columns indicate year of birth and rows are the years respondents are observed in the ACS. Our individual sample contains men born between 1979 and 1983 and women who were born between 1982 and 1986 (highlighted in orange, gray, and green). Our couple sample consists of all identified partnerships in which one partner is a member of the individual sample and the other has an age-birth cohort combination that falls inside the yellow box.

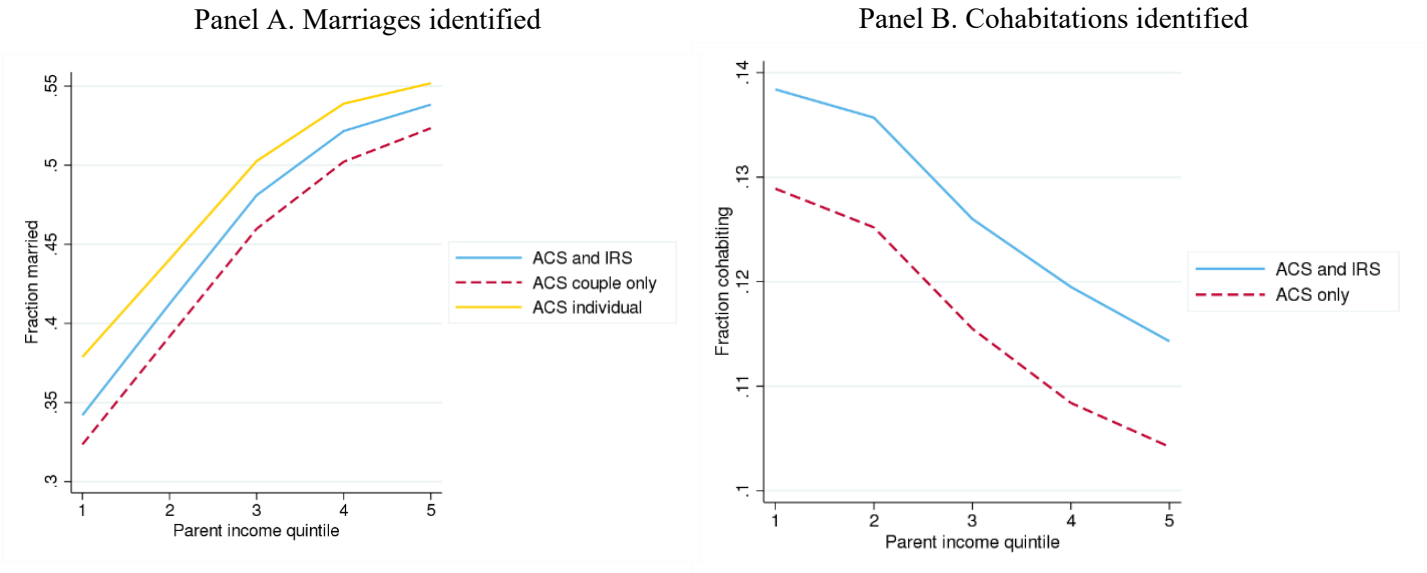


FIGURE A2. ADDITIONAL SPOUSES AND PARTNERS GAINED FROM TAX DATA

Note: This figure plots the fraction of individuals in each quintile of childhood family income by different relationship categorizations. In Panel A, we show the fraction of ACS respondents who report being married (yellow, solid), the fraction of ACS respondents who we can link to a spouse using the ACS (red, dashed), and the fraction of ACS respondents who we can link to their spouse using the ACS and filing status in the IRS 1040s. Panel B shows the fraction of individuals whose unmarried partners can be identified using the ACS (red, dashed) and the fraction of individuals whose unmarried partners can by identified using the ACS or by IRS 1040 filing status.
Source: Individual sample described in Section I.A, which is generated by linking 2011-2019 American Community Survey to 1040 Forms from the Internal Revenue Service.

TABLE A1—INTERGENERATIONAL MOBILITY OF WOMEN'S FAMILY INCOMES

Mobility statistic	White, NH	Black, NH	Hispanic	AIAN, NH	Asian, NH
Panel A. Observed Data					
Pr(move out of Q1)	0.73	0.57	0.72	0.54	0.83
Pr(move from Q1 to Q5)	0.11	0.03	0.08	0.03	0.27
Pr(remain in Q5)	0.39	0.16	0.29	0.22	0.44
Pr(move from Q5 to Q1)	0.08	0.20	0.13	0.17	0.10
Panel B. Counterfactual: All Groups have Black Women's Personal Income					
Pr(move out of Q1)	0.83	0.70	0.79	0.75	0.80
Pr(move from Q1 to Q5)	0.19	0.08	0.15	0.10	0.26
Pr(remain in Q5)	0.53	0.27	0.45	0.42	0.45
Pr(move from Q5 to Q1)	0.06	0.14	0.08	0.08	0.09
Panel C. Counterfactual: All Groups Have Black Women's Partner Income					
Pr(move out of Q1)	0.73	0.78	0.78	0.67	0.86
Pr(move from Q1 to Q5)	0.13	0.13	0.13	0.08	0.34
Pr(remain in Q5)	0.42	0.37	0.38	0.32	0.56
Pr(move from Q5 to Q1)	0.11	0.11	0.11	0.15	0.09

Note: "NH" stands for non-Hispanic; "AIAN" stands for American Indian or Alaskan Native. "Q" refers to quintile. This table maps the distribution of CFI into the distribution of adult family income (AFI), which is the sum of an individual's personal income and their partner's income (set to zero if no partner present). For example, "Pr(move out Q1)" is the conditional probability that a women from the bottom CFI quintile is not in the bottom quintile of AFI. Panel A is the observed data. Panel B (C) replaces observed personal (partner) income for each woman from CFI percentile p with the personal (partner) income of a randomly-selected White non-Hispanic woman also from percentile p. This process is repeated 100 times; the reported mobility statistics are averaged across these 100 counterfactual distributions.

Source: 2011-2019 American Community Survey linked to 1040 Forms from the Internal Revenue Service.